Inverted MAML Card 1: Round 3 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.02, L2=2560, Mom=0.8, Clamp=±70 LSB

1. Round 3 Summary & Cumulative Comparison (R1–R3)

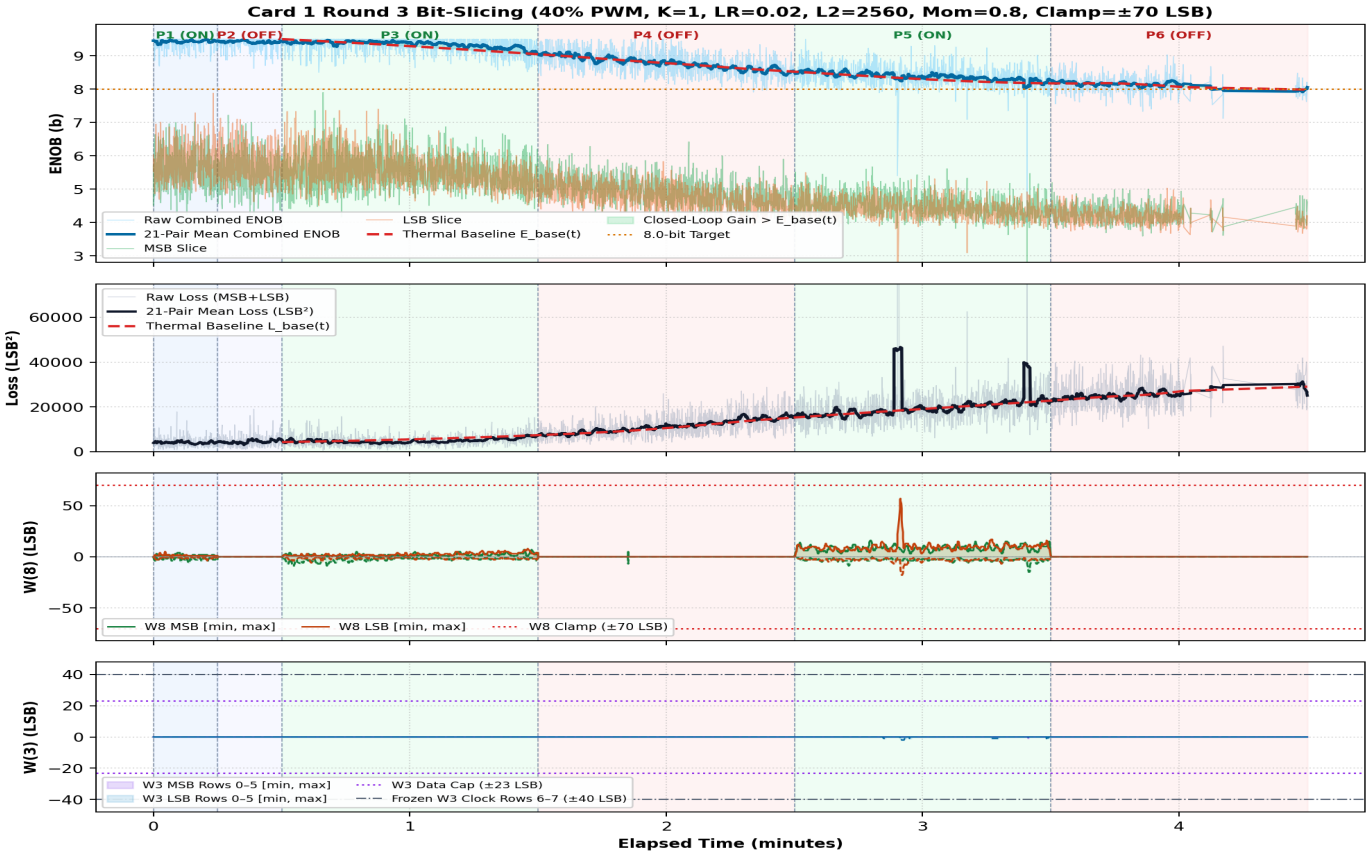
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 3 Detrended Trend Metrics:** Lift vs E_base = **+0.09 b**, Loss Ratio = **1.11x**, Climb = **+0.22 b**, Jitter = **0.26 b**, Recovery = **1.00**, Score = **+0.559**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R1	0.03	1024	0.9	±75	+0.66b	1.57x	+0.74b	0.33b	0.67	+1.957
R2*	0.035	1536	0.85	±75	+0.50b	1.38x	+0.82b	0.29b	0.96	+2.004
R3	0.02	2560	0.8	±70	+0.09b	1.11x	+0.22b	0.26b	1.00	+0.559

2. Phase-by-Phase Telemetry (3,961 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	331	9.43->9.44 (9.43b)	-	4446->3828	[6,5]	0	330
P2	Cold / Regler OFF	310	9.41->9.40 (9.42b)	-	4413->4719	[0,0]	0	0
P3	Warm / Regler ON	1072	9.44->9.26 (9.35b)	+0.12 b	4044->5475	[9,8]	0	1071
P4	Warm / Regler OFF	938	9.02->8.61 (8.79b)	-0.01 b	8070->14121	[7,0]	0	1
P5	Warm / Regler ON	850	8.48->8.28 (8.38b)	+0.07 b	16754->22708	[16,57]	2	850
P6	Warm / Regler OFF	460	8.21->8.09 (8.14b)	+0.02 b	22739->27025	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 3

• **Full 84 W(8) + 96 W(3) Weight Coverage:** Displays the signed `[min, max]` excursion envelopes of all 42 `W(8) MSB`, 42 `W(8) LSB`, and 72 `W(3)` active data synapses (capped at ±23 LSB), plus the 24 frozen `W(3)` clock staircase weights in rows 6–7 (`[-40..+40]` & `[+40..-40]` LSB). Full per-synapse 180-trace trajectories are logged natively from Round 4 onward.

Round 3 W(8) & W(3) Bank Coverage (84 W8 + 96 W3 Synapses; Clamp=±70, M3 Cap=±23)

